# HIGH PERFORMANCE COMPUTING

## Assignment-2

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**BATCH:** 12

### Section 2: Naïve Matrix Multiplication (O(n3) compute)

### Aim

Measure serial performance of a triple-loop matrix multiplication implemented in pure Python, and profile computational hotspots.

Requirements

- Python 3.8+
- Standard library: time, random, cProfile, pstats, io, tracemalloc.

### Code:

```
import random
import time
import cProfile
import pstats
import io
import tracemalloc
# ---------------------------------
# Generate a random n x n matrix
# ---------------------------------
def generate_matrix(n):
    return [[random.random() for _ in range(n)] for _ in range(n)]
# ---------------------------------
```

```
# Naïve matrix multiplication (O(n^3))
# ---------------------------------
def matmul_naive(A, B):
    n = len(A)
    C = [[0.0 for _ in range(n)] for _ in range(n)]
    for i in range(n):
        for j in range(n):
            for k in range(n):
                C[i][j] += A[i][k] * B[k][j]
    return C
# ---------------------------------
# Measure execution time
# ---------------------------------
def run_timing(n):
    A = generate_matrix(n)
    B = generate_matrix(n)
    start = time.perf_counter()
    matmul_naive(A, B)
    end = time.perf_counter()
    print("=== Execution Time ===")
    print(f"Matrix size : {n} x {n}")
    print(f"Time taken  : {end - start:.4f} seconds\n")
# ---------------------------------
# CPU profiling using cProfile
# ---------------------------------
def run_cpu_profile(n):
    A = generate_matrix(n)
    B = generate_matrix(n)
    profiler = cProfile.Profile()
    profiler.enable()
    matmul_naive(A, B)
```

```
    profiler.disable()
    s = io.StringIO()
    stats = pstats.Stats(profiler, stream=s).sort_stats("cumulative")
    stats.print_stats(10)
    print("=== CPU Profiling (Top 10) ===")
    print(s.getvalue())

# ----------------------------------
# Memory profiling using tracemalloc
# ----------------------------------
def run_memory_profile(n):
    A = generate_matrix(n)
    B = generate_matrix(n)
    tracemalloc.start()
    matmul_naive(A, B)
    current, peak = tracemalloc.get_traced_memory()
    tracemalloc.stop()
    print("=== Memory Profiling ===")
    print(f"Current memory usage : {current / 10**6:.2f} MB")
    print(f"Peak memory usage   : {peak / 10**6:.2f} MB\n")
# ----------------------------------
# Main function
# ----------------------------------
def main():
    N = 200   # Change carefully: 100, 200, 300 (higher = much slower)
    print("\nNaïve Matrix Multiplication Performance Analysis\n")
    run_timing(N)
    run_cpu_profile(N)
    run_memory_profile(N)
if __name___== "__main__":
    main()
```

## LOCAL OUTPUT:

```
Naïve Matrix Multiplication Performance Analysis

=== Execution Time ===
Matrix size : 200 x 200
Time taken  : 0.6044 seconds

=== CPU Profiling (Top 10) ===
         4 function calls in 0.636 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.636    0.636    0.636    0.636 C:\Users\ARUN REDDY\Downloads\Ladder-Problem.py:15(matmul_naive)
        1    0.000    0.000    0.001    0.001 C:\Users\ARUN REDDY\Downloads\Ladder-Problem.py:17(<listcomp>)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
        1    0.000    0.000    0.000    0.000 {built-in method builtins.len}


=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage    : 1.29 MB

>>>
```

## GOOGLE COLAB

## CPU:

```
...  Naïve Matrix Multiplication Performance Analysis

     === Execution Time ===
     Matrix size : 200 x 200
     Time taken  : 1.3545 seconds

     === CPU Profiling (Top 10) ===
             355 function calls (353 primitive calls) in 1.146 seconds

        Ordered by: cumulative time
        List reduced from 109 to 10 due to restriction <10>

        ncalls  tottime  percall  cumtime  percall filename:lineno(function)
             2    0.000    0.000    1.140    0.570 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
             2    0.000    0.000    1.129    0.565 /usr/lib/python3.12/selectors.py:451(select)
             2    0.484    0.242    1.129    0.565 {method 'poll' of 'select.epoll' objects}
             1    0.645    0.645    0.645    0.645 /tmp/ipython-input-4270066419.py:15(matmul_naive)
             2    0.000    0.000    0.017    0.008 /usr/lib/python3.12/asyncio/events.py:86(_run)
             2    0.000    0.000    0.017    0.008 {method 'run' of '_contextvars.Context' objects}
             8    0.016    0.002    0.016    0.002 /usr/local/lib/python3.12/dist-packages/zmq/sugar/socket.py:632(send)
             1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
             1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
             1    0.000    0.000    0.011    0.011 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_recv)


     === Memory Profiling ===
     Current memory usage : 0.01 MB
     Peak memory usage    : 1.29 MB
```

## TPU:

```
...  Naïve Matrix Multiplication Performance Analysis

     === Execution Time ===
     Matrix size : 200 x 200
     Time taken  : 0.5977 seconds

     === CPU Profiling (Top 10) ===
             490 function calls (486 primitive calls) in 0.540 seconds

        Ordered by: cumulative time
        List reduced from 123 to 10 due to restriction <10>

        ncalls  tottime  percall  cumtime  percall filename:lineno(function)
             3    0.000    0.000    0.540    0.180 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
             3    0.000    0.000    0.524    0.175 /usr/lib/python3.12/selectors.py:451(select)
             1    0.318    0.318    0.318    0.318 /tmp/ipython-input-4270066419.py:15(matmul_naive)
             2    0.205    0.103    0.205    0.103 {method 'poll' of 'select.epoll' objects}
             3    0.000    0.000    0.016    0.005 /usr/lib/python3.12/asyncio/events.py:86(_run)
             3    0.000    0.000    0.016    0.005 {method 'run' of '_contextvars.Context' objects}
             8    0.015    0.002    0.015    0.002 /usr/local/lib/python3.12/dist-packages/zmq/sugar/socket.py:632(send)
             2    0.000    0.000    0.011    0.005 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
             2    0.000    0.000    0.011    0.005 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
             2    0.000    0.000    0.010    0.005 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_recv)


     === Memory Profiling ===
     Current memory usage : 0.01 MB
     Peak memory usage    : 1.29 MB
```

## GPU:

```
Naïve Matrix Multiplication Performance Analysis

=== Execution Time ===
Matrix size : 200 x 200
Time taken  : 0.8500 seconds

=== CPU Profiling (Top 10) ===
         491 function calls (486 primitive calls) in 0.778 seconds

   Ordered by: cumulative time
   List reduced from 123 to 10 due to restriction <10>

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
      4/3    0.000    0.000    0.777    0.259 /usr/lib/python3.12/asyncio/base_events.py:1922(_run_once)
        3    0.001    0.000    0.761    0.254 /usr/lib/python3.12/selectors.py:451(select)
        1    0.555    0.555    0.555    0.555 /tmp/ipython-input-4270066419.py:15(matmul_naive)
        2    0.205    0.103    0.205    0.103 {method 'poll' of 'select.epoll' objects}
        3    0.000    0.000    0.016    0.005 /usr/lib/python3.12/asyncio/events.py:86(_run)
        3    0.000    0.000    0.016    0.005 {method 'run' of '_contextvars.Context' objects}
        2    0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/tornado/platform/asyncio.py:206(_handle_events)
        2    0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:574(_handle_events)
        2    0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:615(_handle_recv)
        2    0.000    0.000    0.016    0.008 /usr/local/lib/python3.12/dist-packages/zmq/eventloop/zmqstream.py:547(_run_callback)


=== Memory Profiling ===
Current memory usage : 0.01 MB
Peak memory usage    : 1.29 MB
```